MILITARY SPECIFICATION

BOXES, CONSOLIDATION, FIBERBOARD AND POLYOLEFIN

This specification is approved for use by the Air Force Acquisition Logistics Division, Air Force Packaging Evaluation Agency, AFALD/PTP, Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for weather-resistant, high stacking strength, double-wall, and triple-wall corrugated fiberboard, and polyolefin boxes of special design and characteristics (see 6.1).

1.2 Classification. Consolidation boxes described herein shall be of the following types, styles, and sizes as specified in table I (see 3.4.1 and 6.2).

1.2.1 Types

Type I - Double-wall corrugated fiberboard
Type II - Triple-wall corrugated fiberboard
Type III - Polyolefin material

1.2.2 Styles

Style S - Sleeve (tube body) with double cover
Style T - Half-slotted body with cover (transporter)
Style U - One-piece palletized container with cover

1.3 Part Number. Specification part numbers for items described in this specification will be formulated as shown in section 6.3.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal:

QQ-S-781	-Strapping, Steel, Flat and Seals
PPP-B-636	-Box, Fiberboard
PPP-B-638	-Boxes, Caps, Liners, and Sleeves Fiberboard, Knocked-down, Flat; Packing of
PPP-B-640	-Boxes, Fiberboard, Corrugated, Triple-wall
PPP-B-1364	-Box, Corrugated Fiberboard, High Strength, Weather-resistant, Double-wall
PPP-S-760	-Strapping, Nonmetallic, (and connectors)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (Air Force Packaging Evaluation Agency, AFALD/PTP, Wright-Patterson AFB OH 45433), by using the self-addressed standardization document improvement proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Federal:

FED TEST METHOD STD NO. 101	-Preservation, Packaging, and Packing Materials: Test Procedures
FED-STD-123	-Marking for Shipment (Civil Agencies)

Military:

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-Marking for Shipment and Storage

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies shall be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E4 - Tentative Methods of Verification of Testing Machines

D641 - Conditioning Paperboard, Fiberboard and Paperboard Containers for Testing

D642 - Compression Test for Shipping Containers

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.2 Materials.

3.2.1 Fiberboard.

3.2.1.1 Fiberboard for type I boxes. The fiberboard shall be corrugated, double-wall PPP-B-1364 high strength, weather-resistant.

3.2.1.2 Fiberboard for type II boxes. The fiberboard shall be corrugated, triple-wall PPP-B-640 class weather-resistant.

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3.2.2 Polyolefin for type III boxes. The polyolefin material shall pass the tests specified in 4.5 as applicable for the container requirements.

3.2.3 Staples for Fiberboard Box Body or sleeve. The wire for staples (preformed or machine stitched) shall be flat minimum size 0.103 inch (2.616 mm) wide, by 0.020 inch (0.580 mm) thick. Tolerances on the width shall be plus 0.002 inch (0.051 mm) or minus 0.008 inch (0.203 mm). Tolerances on thickness shall be plus unlimited or minus 0.001 inch (0.025 mm). The crown width of the staples or stitch shall be 5/8 inch (15.875 mm) with a tolerance of minus 3/16 inch (4.762 mm) or plus 1/8 inch (3.175 mm), and a leg length sufficient to provide a 1/4 inch (6.350 mm) clinch. The wire shall have a commercially applied coating or plating such as zinc (galvanized) or copper wash.

3.2.3.1 Staples for fiberboard covers or caps. Preformed staples shall be wide crown 1 1/4 inch (31.750 mm) minimum outside dimension, made of wire 0.088 inch (2.235 mm) minimum width, by 0.0545 inch (1.384 mm) minimum thickness, and a leg length sufficient to provide a 1/4 to 3/8 inch (6.350 to 9.525 mm) clinch. Machine formed stitches shall be made of flat wire, minimum size 0.103 inch (2.616 mm) wide, by 0.020 inch (0.580 mm) thick. Tolerances on width shall be plus 0.002 inch (0.051 mm) or minus 0.008 inch (0.203 mm). Tolerances on thicknesses shall be plus unlimited or minus 0.001 inch (0.025 mm). The crown width shall be a minimum 5/8 inch (15.875 mm) with a tolerance of minus 3/16 inch (4.762 mm) or plus 1/8 inch (3.175 mm) and sufficient to provide a 1/4 inch (6.350 mm) clinch.

3.3 Performance Requirements.

3.3.1 Bending. The fiberboard for type I or type II boxes shall show no continuous visual surface break of the outer component ply or facing and shall not split completely through the outer facing when folded as specified in 4.5.2.

3.3.2 Compression. The performance of the type I, II or III boxes shall be such that the top-to-bottom compression strength value of the boxes fabricated in accordance with paragraph 3.4 shall be equal to or better than the value specified in table II (see 4.3.3).

3.3.3 Edgewise-Drop. The performance of the type I, II or III boxes shall be such that the edgewise-drop (rotational) test as specified in 4.5.4.3, shall show no functional or physical damage to the boxes.

3.3.4 Cornerwise-Drop. The performance of the type I, II or III boxes shall be such that the cornerwise-drop (rotational) test as specified in 4.5.4.4, shall show no functional or physical damage to the boxes.

3.3.5 Vibration. The performance of the type III boxes shall be such that the vibration test, as specified in 4.5.4.5, will not cause failure to the boxes.

3.3.6 Pendulum-impact. The performance of the type III boxes shall be such that the pendulum-impact test as specified in 4.5.4.6 shall show no functional or physical damage to the boxes.

3.3.7 Temperature. The temperature used as specified in 4.5.3.1.3 for the performance of the type III boxes shall show no functional or physical damage to the boxes.

3.4 Construction.

3.4.1 Sizes. Unless otherwise specified, sizes shall be as specified in table I. When sizes other than those listed in table I are required, they shall meet all other requirements of this document (see 6.2).

3.4.2 Dimensions. The dimensions of the boxes shall be the inside measurements and shall be given in sequence of length, width, and depth (see Figure 1 and 2). A tolerance of +1/4 inch (6.350 mm) shall be permitted in the dimensions of the assembled fiberboard boxes and ±1/8 inch (3.175 mm) in the assembled plastic boxes.

3.4.3 Gross weight limitation. The gross weight limitation for each type, style and size of box shall be as specified in table II. For test procedures ±5 percent shall be permitted.

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TABLE I. Dimension specifications

SIZES	TYPES	STYLES	DIMENSION (INSIDE) INCHES/MILLIMETERS
Sixteenth	I	S	23 x 18-1/2 x 15 (584.2 x 469.9 x 381.0)
	III	U	23 x 18-1/2 x 15 (584.2 x 469.9 x 381.0)
Eighth	I	S	23 x 18-1/2 x 29 (584.2 x 469.9 x 736.6)
	II	S	22-1/2 x 18 x 28-1/2 (571.5 x 457.2 x 723.9)
	III	U	23 x 18-1/2 x 29 (584.2 x 469.9 x 736.6)
Quarter	I	S	40 x 23 x 29 (1016.0 x 584.2 x 736.6)
	II	S	39-1/2 x 22-1/2 x 28-1/2 (1003.3 x 571.5 x 723.9)
	III	U	40 x 23 x 29 (1016.0 x 584.2 x 736.6)
Half	I	S	48 x 40 x 28-1/2 (1219.2 x 1016.0 x 723.9)
	II	S&T	47-1/2 x 39-1/2 x 28 (1206.5 x 1003.3 x 711.2)
	III	U	48 x 40 x 28-1/2 (1219.2 x 1016.0 x 723.9)
Full	II	S&T	83 x 49 x 28 (2108.2 x 1244.9 x 711.2)
	III	U	83-1/2 x 49-1/2 x 28-1/2 (2120.9 x 1257.3 x 723.9)

3.4.4 Sleeve for style S box body. The sleeve shall be fabricated from one piece of scored fiberboard except for half or full size boxes which may be fabricated with two body (manufacturer's) joints on diagonally opposite corners. Either fabrication must meet all test requirements. The direction of the corrugations shall be perpendicular to the open faces. Body (manufacturer's) joint shall be minimum of 2.0 inches (50.8 mm) wide and compressed stitched (see Figure 1, detail B). Prior to stapling, the corrugations in the 2.0 inch (50.8 mm) wide strip shall be completely crushed on both the body and the strip. The metal staples shall be placed on a slant across the strip and shall be spaced not more than 1.0 inch (25.4 mm) apart; the 1.0 inch (25.4 mm) measurement shall be from the lower tip of one staple to the top tip of the staple directly below (see Figure 1, detail A).

3.4.5 Half slotted body for style T boxes. The fabrication may be made with two body (manufacturer's) joints on diagonally opposite corners and shall be in accordance with the requirements of PPP-B-636, style RSC, except it shall be half slotted and the body (manufacturer's) joint shall be a minimum of 2.0 inches (50.8 mm) wide and compressed stitched (see Figure 1, detail B). The fiberboard used shall be as specified herein. The slots shall have a minimum width of 3/8 inch (9.525 mm). Prior to stapling, the corrugations in the 2.0 inch (50.8 mm) wide strip shall be completely crushed on both the body and the strip. The metal staples shall be placed on a slant across the strip and shall be spaced not more than 1.0 inch (25.4 mm) apart; the 1.0 inch (25.4 mm) measurement shall be from the lower tip of one staple to the top tip of the staple directly below (see Figure 1, detail A). The directions of the flutes shall be vertical in the sides and ends of the box.

3.4.6 Body for style U boxes. The body shall be a one-piece construction. The method of fabrication is not fixed but is to be within limits imposed by a good manufacturing process. Bodies, when empty, shall be capable of nesting 75 percent of their height to conserve space (see Figure 3).

3.4.7 Covers. The covers for the style S and T boxes shall be made from a single piece corrugated fiberboard and cut so as to form a cover having four flanges. Corrugation direction may be either parallel or perpendicular to the length dimension of the cover. The flanges shall have a minimum width of 4.0 inches (101.6 mm) and shall be so arranged that when assembled the flaps shall fasten to the outside of the flanges having the longest dimension. The corrugations in both the flap and flange overlapping areas shall be completely crush rolled to facilitate stapling with 3/4 inch (19.050 mm) leg staples. In addition to crushing areas where flaps overlap (for stitching purposes), there shall be an additional crush 1.0 inch (25.4 mm) wide on all outer edges of the flanges for style S boxes. Unless otherwise specified (see 6.2), covers being prepared for delivery shall not be stapled but shall be crushed, scored, and slotted (see Figure 4). Cover for the style U boxes shall be one piece, constructed to form a flange having a minimum width of 2-1/2 inches (63.50 mm) (see Figure 3).

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3.5 Marking.

3.5.1 Boxmaker's certificate. Each fiberboard box shall bear the certificate of the boxmaker in compliance with the requirements of the carrier's freight classification rules.

3.5.2 Special Marking. Special marking (on the fiberboard boxes) shall be in 1/2 inch (12.700 mm) bold face print. Each cover and sleeve or box body shall be imprinted with the National Stock Number, style, and size. Markings shall be placed in the lower right corner of each panel cover and in the lower right corner of the length dimension of the sleeve, allowing 7.0 inches (177.8 mm) from the bottom of the line of the imprint to the bottom of the sleeve.

3.5.3 Embossing. The style U boxes shall be embossed with "U.S. Air Force Reusable Container/Cover." Markings shall be in 1.0 inch (25.4 mm) minimum bold face letters. Markings shall be placed in the lower right corner of the length dimension of the box approximately 7.0 inches (177.8 mm) from the bottom and in the lower right hand corner of the length dimension of the cover top.

3.6 Workmanship. Boxes manufactured in accordance with the requirements of this specification shall be free from imperfections that may affect their serviceability. Dimensions of the fiberboard boxmaker's blank shall be accurately cut and scored to permit 135° bending and shall be so slotted that the assembled box meets the dimensional requirements without undue binding or excessive gap in the fit between the covers and the sleeve.

3.7 Filled boxes. When boxes are furnished as filled boxes, the closure and strapping of the boxes shall be in accordance with paragraph 4.5.4.2.2 and Figure 7.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of boxes shall be classified as follows:

a. First article test (see 3.1, 4.3.1 and 4.3.3).

b. Quality conformance test (see 4.4).

4.3 First article tests.

4.3.1 First article test samples. First article test samples shall consist of one box of each type and style representative of the material specified for that type of box, production design, and construction upon which approval is desired. First article tests of boxes shall be conducted as specified by the procuring activity. Samples shall be appropriately identified with the manufacturer's part number and any additional information required by the procuring activity. Unless otherwise specified, the box selected for first article testing shall be the largest box within a given type and style for which procurement is being effected. Boxes shall be tested and approved by the procuring activity before production is commenced.

4.3.2 First article tests. First article tests shall be repeated in the event a change in the manufacturing process is made.

4.3.3 Test. The first article test shall consist of all the tests and examinations specified herein and shall apply to the test samples of 4.3.1. The gross weights, compression test (pounds/kilograms), and drop tests for the various sizes of boxes shall be as specified in table II.

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4.4 Quality conformance tests. These tests shall consist of the examination of the product as specified in 4.4.2 and the performance tests specified in 4.5.2, 4.5.3 and 4.5.4.

4.4.1 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated. For the purpose of sampling, an inspection lot for examination and test shall consist of all materials of the same type and style submitted for delivery at one time. When multiple sizes of any one type and style are specified, the inspection lot may consist of all sizes presented at one time in which case sampling shall be in proportion or ratio to the respective sizes in the lot. The acceptance quality level (AQL) shall be applicable to the entire lot regardless of defects as they may occur in any one or more size of box.

4.4.2 Examination.

4.4.2.1 Examination of the end item. Examination of the end item shall be made in accordance with the classification of defects, inspection levels, and acceptance quality levels (AQLs) set forth below. The lot size, for the purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in unit of boxes. For examination the lot and size shall be expressed in units of shipping containers, boxes, pallets, bundles, or crates. The sample for this unit shall be one completely fabricated box, knocked-down, partly or completely assembled.

EXAMINE	DEFECTS
MATERIAL:	
Type of board or polyolefin	Not as specified.
Condition	Not new, unduly dirty, stained, or scuffed.
	Fiberboard does not set up properly due to improper scoring or slotting. Outer ply or facing of fiberboard split completely through when folded.
Metal fasteners	Not as specified. Cracked or rusted.
CONSTRUCTION:	
Body	Not fabricated from one or two pieces of fiberboard or one-piece polyolefin as prescribed.
	Not type specified.
	Fiberboard joint along one or two edges of the four edges not perpendicular to the open faces.
Cover	Not constructed of one-piece fiberboard or polyolefin material.
	Fiberboard corrugations not completely crush-rolled on both the flap and flap overlapping area of flange.
	Does not provide snug fit between cover and sleeve.
	Fiberboard flaps not overlapped on the outside of the flange with the longest dimension.
	Dimensions not as specified.
Scoring	Poor scoring on fiberboard, making it excessively difficult to fold into required position. Outer ply or facing of fiberboard split completely through when folded.
Marking	Missing, not legible, incomplete, incorrect, of improper size, or not located where specified.
Manufacturer's certification of compliance	Missing, not legible, incomplete, incorrect, or not located where specified on fiberboard boxes.

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EXAMINE	DEFECTS
Body joints	Overlapped areas of fiberboard do not consist of a 2.0 inch (50.8 millimeter) crush-rolled strip of entire length of manufacturer's joint.
Metal fasteners on fiberboard boxes	Does not pass completely through all pieces to be fastened. Not placed on slant. One or more required fasteners not completely within overlap area. More than 1.0 inch (25.4 millimeter) apart.
Dimensions	All measurements shall be the inside measurement of the body. Measure length from one end of body to the other end of the side opposite joint. Measure width along end opposite joint from one side to the other side. Determine height by measuring the body perpendicular to the open faces.

4.4.2.1.1 Examination of end item for delivery. An examination shall be made to determine compliance with packing and marking requirements of section 5 and the contract as applicable. Examination shall be performed as specified in PPP-B-638.

4.4.2.1.2 Inspection levels. Inspection levels and acceptable quality levels for the purpose of determining the sample size and acceptable quality level (AQL) expressed in defects per 100 units shall be as indicated in table III.

4.5 Test procedures.

4.5.1 Conditioning.

4.5.1.1 Conditioning for dry tests. Fiberboard samples and boxes for all tests shall be conditioned in accordance with ASTM method D641.

4.5.1.2 Conditioning for type III container tests. The compression, pendulum-impact, and vibration tests shall be conducted at ambient temperature. The cornerwise and edgewise drops shall be conducted at -20°F ± 5°F (-7°C ± 3°C) and 140°F ± 5°F (60°C ± 3°C). The superimposed load temperature test shall be conducted at 160°F ± 5°F (71°C ± 3°C) (see 4.5.3.1.3).

4.5.2 Bending.

4.5.2.1 Type I and type II boxes. One fiberboard specimen not less than 12 inches (304.8 mm) by 12 inches (304.8 mm) shall be cut from a sound, undamaged, and unprinted area of each sample box in a lot. The specimens shall be conditioned in accordance with paragraph 4.5.1.1. The specimens shall be scored along and across the machine direction, or corrugations, using scoring devices having the proper profiles for the type of fiberboard to be tested. There shall be no surface breaks in the board when folded 180° along the score line parallel to the flute direction and 90° along the score line perpendicular to the flute direction (see 3.3.1).

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TABLE II. Compression and Drop Tests

SIZE	TYPE	STYLE	GROSS WT OF BOXES*		COMPRESSION TEST		EDGEWISE (ROTATIONAL) DROP TEST ** HEIGHT OF DROP		CORNERWISE (ROTATIONAL) DROP TEST ** HEIGHT OF DROP	
			POUNDS	KILOGRAMS	POUNDS	KILOGRAMS	INCHES	MILLIMETERS	INCHES	MILLIMETERS
SIXTEENTH	I	S	200	90.70	2500	1133.60	24	609.60	24	609.60
	III	U	200	90.70	2500	1133.60	24	609.60	24	609.60
EIGHTH	I	S	500	226.80	3000	1360.50	24	609.60	24	609.60
	II	S	500	226.80	5000	2267.50	24	609.60	24	609.60
	III	U	600	272.10	5000	2267.50	24	609.60	24	609.60
QUARTER	I	S	500	226.80	3000	1360.50	24	609.60	24	609.60
	II	S	500	226.80	5000	2267.50	24	609.60	24	609.60
	III	U	600	272.10	5000	2267.50	24	609.60	24	609.60
HALF	I	S	500	226.80	3000	1360.50	24	609.60	24	609.60
	II	S & T	1000	453.50	5000	2267.50	24	609.60	24	609.60
	III	U	1200	544.20	5000	2267.50	24	609.60	24	609.60
FULL	II	S & T	1000	453.50	5000	2267.50	24	609.60	24	609.60
	III	U	1200	544.20	5000	2267.50	24	609.60	24	609.60

* Gross weight shall be as specified above with a tolerance of ±5 percent.

** If the size of the box and the location of the center of gravity are such that this drop cannot be made from the prescribed height, the greatest attainable height shall be substituted.

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4.5.3 Compression strength test of boxes.

4.5.3.1 Test apparatus. The apparatus for making the compression tests shall consist of a compression machine capable of sufficient capacity to meet the tests. (The use of supplementary pressure plates between the platens of the compression tester shall be permitted only with the approval of the procuring activity.) The test apparatus shall be calibrated in accordance with the tentative methods of verification of testing machines (ASTM method E-4).

4.5.3.1.1 Compression test. Boxes shall be conditioned in accordance with paragraph 4.5.1.1 and 4.5.1.2 and shall be tested for top-to-bottom compression strength according to ASTM method D642. The test shall be continued until it is clearly evident that the acceptable compression load has been attained. Each assembled empty box shall be tested to the compression loads (pounds/kilograms), as specified in Table II. The maximum deflection shall not be more than 2.0 inches (50.8 mm) for type I and type II boxes and 1.0 inch (25.4 mm) for type III boxes.

4.5.3.1.2 Static test. In the event the test apparatus, as specified in 4.5.3.1, is not available, a static load compression test is authorized. Boxes shall be assembled empty and conditioned in accordance with paragraph 4.5.1.1 and 4.5.1.2. A test load, equal to 70 percent of the compression test (pounds/kilograms) specified in table II, shall be uniformly distributed over the top of the test container. The load shall be permitted to stand in a static position for a period of not more than 5 minutes. The maximum deflection shall not be more than 2.0 inches (50.8 mm) for type I and type II boxes and 1.0 inch (25.4 mm) for type III boxes during the 5 minute period. If deflection beyond 2.0 inches (50.8 mm) or 1.0 inch (25.4 mm) occurs during the 5 minute period, the box fails to meet the static test requirements.

4.5.3.1.3 Temperature test type III box. An assembled empty box with a superimposed load of 25 percent of the compression load (pounds/kilograms) table II shall be tested at 160°F ±5°F (71°C ±3°C) for 24 hours. The maximum deflection shall not be more than 1.0 inch (25.4 mm) during a 24-hour period. If deflection beyond 1.0 inch (25.4 mm) occurs during the 24-hour period, the box fails to meet the temperature test requirements.

4.5.4 Rough handling tests.

4.5.4.1 Preparation of box for rough handling tests. Each box shall be assembled complete with cover/cap, base and strapping in accordance with paragraph 3.4 and figures 1, 2, 3 and 7. The assembled boxes shall be tested with a dummy load as specified in 4.5.4.2.

4.5.4.2 Dummy load. The prototype box(es) shall be packed with an assortment of various sizes of boxes. Approximately 90 percent of the inside of the test box area shall be utilized. The assortment of boxes shall be fabricated from any type and style of PPP-B-636 fiberboard material. The assorted sizes of boxes shall be filled with ballast (sand or gravel) material having the proper density to meet the gross weight requirements for the box(es) being tested (see table II). After the interior boxes are filled with ballast, the weighted box(es) shall be uniformly distributed over the entire inside of the test box area. The top cover shall then be placed over the top portion of the box to form an assembled box.

4.5.4.2.1 Strapping. After the boxes have been assembled and packed, each box shall be strapped. Strapping material shall be 5/8 inch (15.875 mm) wide and shall have a minimum thickness of 0.020 inch (0.508 mm) conforming to QQ-S-781 or PPP-S-760 of comparable tensile strength. Strapping shall be applied first in the lengthwise position and then it shall be applied in the girthwise position (see figure 7).

4.5.4.2.2 Strapping position. Strapping shall be applied to the applicable size of boxes in the following positions (see figure 7):

Sixteenth and Eighth Sizes - One band centered lengthwise and one band centered girthwise.

Quarter Size - One band lengthwise in the center of the container and two bands girthwise spaced one-third of the distance between the ends of the container.

Half Size - Two bands lengthwise spaced one-third of the distance between sides of the container and two bands girthwise spaced one-third of the distance between ends of the container.

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Full Size - Two bands lengthwise spaced approximately one-third the distance of the container width dimension; three bands girthwise, one band spaced approximately 18 inches (457.2 mm) from one end, one spaced approximately 18 inches(457.2 mm) from the other end, and one band approximately in the center of the container.

4.5.4.3 Edgewise-drop (rotational) test. The box, with the applicable gross weight specified in table II, shall be tested according to FTMS No. 101, method 5008. The height of the drop shall be as specified in table II. Type III containers shall be conditioned in accordance with 4.5.1.2 prior to this test.

4.5.4.4 Cornerwise-drop (rotational) test. The box with the applicable gross weight specified in table II, shall be tested according to FTMS No. 101, method 5005. The height of the drop shall be as specified in table II. Type III containers shall be conditioned in accordance with 4.5.1.2. When the proportions of width and height of the packed box are such as to cause instability in the cornerwise-drop (rotational) tests, edgewise-drop (rotational) tests shall be substituted for the cornerwise-drop (rotational) tests. In such cases, two additional edgewise-drop (rotational) tests shall be conducted on each end of the box.

4.5.4.5 Vibration (repetitive shock) test type III box. The box with the applicable weight specified in table II shall be tested according to FTMS No. 101, method 5019 and 4.5.1.2.

4.5.4.6 Pendulum-impact test type III box. The box with the applicable weight specified in table II shall be tested according to FTMS No. 101, method 5012 and 4.5.1.2.

4.5.5 Cause for rejection. At the conclusion of the tests, there shall be no visible ruptures in the exterior box panels (covers or body). Rupture is defined as a breaking through of the plies of fiberboard or the polyolefin material to the extent that the interior containers may be seen or the break would allow the spilling of contents.

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TABLE III. Test Requirements.

CHARACTERISTIC	SPEC REFERENCE REQUIREMENT	SPEC REFERENCE TEST METHOD	REQUIREMENTS APPLICABLE TO: INDIVIDUAL UNIT	REQUIREMENTS APPLICABLE TO: LOT AVERAGE	NUMBER DETERMINATIONS PER SAMPLE UNIT	RESULTS REPORTED AS: PASS OR FAIL	RESULTS REPORTED AS: NUMERICALLY TO NEAREST	INSPECTION LEVEL	AQL
Examination of End Product	-----	4.4.2.1	X	---	1	X	----	S-2	2.5
Bending Type I & II	3.3.1	4.5.2.1	X	---	1	X	---	S-2	2.5
Compression Type I, II and III	3.3.2	4.5.3	X	---	1	X	pound/ kilogram	S-2	2.5
Edgewise Drop (Rotational test) Type I, II and III	3.3.3	4.5.4.3	X	---	Once to each end of box	X	---	S-2	2.5
Cornerwise Drop (Rotational test) Type I, II and III	3.3.4	4.5.4.4	X	---	Once to each of two diagonally opposite corners	X	---	S-2	2.5
Vibration (Repetitive Shock) Test Type III	3.3.5	4.5.4.5	X	---	1	X	---	S-2	2.5
Pendulum-Impact Test Type III	3.3.6	4.5.4.6	X	---	One impact to each side and each end	X	---	S-2	2.5
Temperature Test Type III	3.3.7	4.5.3.1.3	X	---	1	X	---	S-2	2.5

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5. PACKAGING.

5.1 Preservation-packaging. Not applicable.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A. The style S and T boxes shall be packed knocked down in accordance with PPP-B-638 as specified therein for overseas shipments. Style U boxes shall be nested, covers stacked and strapped (see Figure 3).

5.2.2 Commercial Packing. The system shall be packaged in accordance with normal commercial practice and packed in fiberboard boxes that will assure acceptance by common carrier and provide protection against loss and damage during shipment, handling, and storage. The shipping container shall be in compliance with the National Motor Freight Classification and the Uniform Freight Classification.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, marking of the exterior container shall be in accordance with MIL-STD-129.

5.3.2 Commercial marking. Marking shall be in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. Boxes covered by this specification are intended for domestic and overseas consolidation shipments of aircraft parts, general supplies, and equipment. Large boxes shall be used by Air Force activities to curtail or eliminate use of high tare weight containers such as CONEX.

6.2 Ordering data. Procurement documents should specify the following.

a. Title, number, and date of this specification.

b. Size, type, and style required (see 1.2.1, 1.2.2, and 3.4.1).

c. Special markings required (see 3.5 and 5.3).

d. Quantity required.

e. Use of edge protector required with metallic strapping (see 30.4).

f. Specify level A or commercial packing (see 5.2).

g. Whether covers should be stapled (see 3.4.7).

h. When first article test sample is required (see 3.1, 4.3 and 6.2.1).

6.2.1 First article test sample. One container of each type and style shall be required as a first article sample, to be subjected to the first article tests, for determination of compliance with the requirements of this specification. The point of inspection for these tests shall be specified by the contract.

6.2.2 Environmental. Refer to material specifications or preparing activity for recommended disposability methods.

6.3 Definitive specification part number. The specification part number is a definitive part number which will be formulated to identify each item covered by this specification. The part number will be formulated by selecting from the requirement or from the options that are available in this specification as follows:

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M38721-X X X

Military specification number
Type designator code (see table IV, 6.3.1)
Style designator code (see table V, 6.3.2)
Size designator code (see table VI, 6.3.3)

6.3.1 Type designator code. Type shall be designated by code as specified in table IV.

TABLE IV TYPE CODE.

Codes	Types	Remarks
A	I	Double-wall corrugated fiberboard
B	II	Triple-wall corrugated fiberboard
C	III	Polyolefin material

6.3.2 Style designator code. Style shall be designated by code as specified in table V.

TABLE V STYLE CODE

Style Code	Remarks
S	Sleeve (tube body) with double cover
T	Half-slotted body with cover (transporter)
U	One-piece palletized container with cover

6.3.3 Size designator code. Size shall be designated by code as specified in table VI.

TABLE VI SIZE CODE.

Codes	Sizes	Types	Styles	Dimension(inside) Inches/Millimeters
A	1/16	I	S	23 x 18 1/2 x 15 (584.2 x 469.9 x 381.0)
A	1/16	III	U	23 x 18 1/2 x 15 (584.2 x 469.9 x 381.0)
B	1/8	I	S	23 x 18 1/2 x 29 (584.2 x 469.9 x 736.6)
C	1/8	II	S	22 1/2 x 18 x 28 1/2 (571.5 x 475.2 x 723.9)
B	1/8	III	U	23 x 18 1/8 x 29 (584.2 x 469.9 x 736.6)
D	1/4	I	S	40 x 23 x 29 (1016.0 x 584.2 x 736.6)
E	1/4	II	S	39 1/2 x 22 1/2 x 28 1/2 (1003.3 x 571.5 x 723.9)
D	1/4	III	U	40 x 23 x 29 (1003.3 x 584.2 x 736.6)
F	1/2	I	S	48 x 40 x 28 1/2 (1219.2 x 1016.0 x 723.9)
G	1/2	II	S&T	47 1/2 x 39 1/2 x 28 (1206.5 x 1003.3 x 711.2)
F	1/2	III	U	48 x 40 x 28 1/2 (1219.2 x 1016.0 x 723.9)
H	Full	II	S&T	83 x 49 x 28 (2108.2 x 1244.9 x 711.2)
I	Full	III	U	83 1/2 x 49 1/2 x 28 1/2 (2120.9 x 1257.3 x 723.9)

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TABLE VII. SAMPLE PART NUMBER	
CHARACTERISTICS	PART NUMBER
Box Shipping, Type I (Sixteenth), 23 x 18 1/2 x 15 (584.2 x 469.9 x 331.0)	M38721ASA
Box Shipping, Type III (Sixteenth), 23 x 18 1/2 x 15 (584.2 x 469.9 x 331.0)	M38721CUA
Box Shipping, Type I (Eighth), 23 x 18 1/2 x 29 (584.2 x 469.9 x 736.6)	M38721ASB
Box Shipping, Type II (Eighth), 22 1/2 x 18 x 28 1/2 (571.5 x 475.2 x 723.9)	M38721BSC

Custodian:
Air Force - 69

Review activities:
Air Force - 43, 99
Army - SM

User activity:
Air Force - 70

Preparing activity:
Air Force - 69

Project No. 8115-F418

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APPENDIX

10. SCOPE

10.1 This appendix covers requirements for using the box(es) as a shipping container for packing Air Force cargo for movement to worldwide destinations.

20 APPLICABLE DOCUMENTS

20.1 The following specifications of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal:

FF-N-105	Nails, Wire, Brads, and Staples
NN-P-530	Plywood, Flat Panel

STANDARDS

Military:

MIL-STD-731	Quality of Wood Members for Containers and Pallets

30 REQUIREMENTS

30.1 Materials.

30.1.1 Plywood. Plywood used shall be in accordance with NN-P-530.

30.1.2 Lumber. Lumber used for the blocks and decking shall be in accordance with MIL-STD-731.

30.1.3 Nails. Nails used in the base and to fasten the blocks to the base frame shall conform to FF-N-105, type 2, style 8, and shall be 12d. Nail points that protrude shall be clinched.

30.1.3.1 Nails. Nails used for the alternate base to fasten the blocks to the plywood deck shall conform to FF-N-105, type 2, style 18 and shall be **2 inches** (50.8 mm) in length (see figure 6).

30.2 Four-way entry handling base. Unless otherwise specified, size shall be as specified in table VIII, and construction of the handling base shall be as shown on figures 5 and 6. This base consists of blocks, stringers, and top deck boards or a full plywood deck which are held together by nails.

30.2.1 Blocks. All blocks shall be approximately 6 inches (152.4 mm) wide by 6 inches (152.4 mm) long by 2 inches (50.8 mm) thick (nominal). For half and full size boxes, blocks shall be inset 2 inches (50.8 mm) from the ends of the base.

30.2.2 Decking. Lumber used for decking shall be 1 x 6 inches (25.4 x 152.4 mm) (nominal).

30.2.3 Stringer. Stringers shall be constructed of either 3/8 or 1/2 inch (9.525 or 12.700 mm) plywood or 1 x 6 (25.4 x 152.4 mm) (nominal) lumber.

30.3 Alternate base. Unless otherwise specified, size shall be as specified in table VIII and constructed as shown on figure 6. This base consists of blocks and a full plywood deck which are held together with nails. This handling base is to be used primarily for the eighth and quarter size boxes.

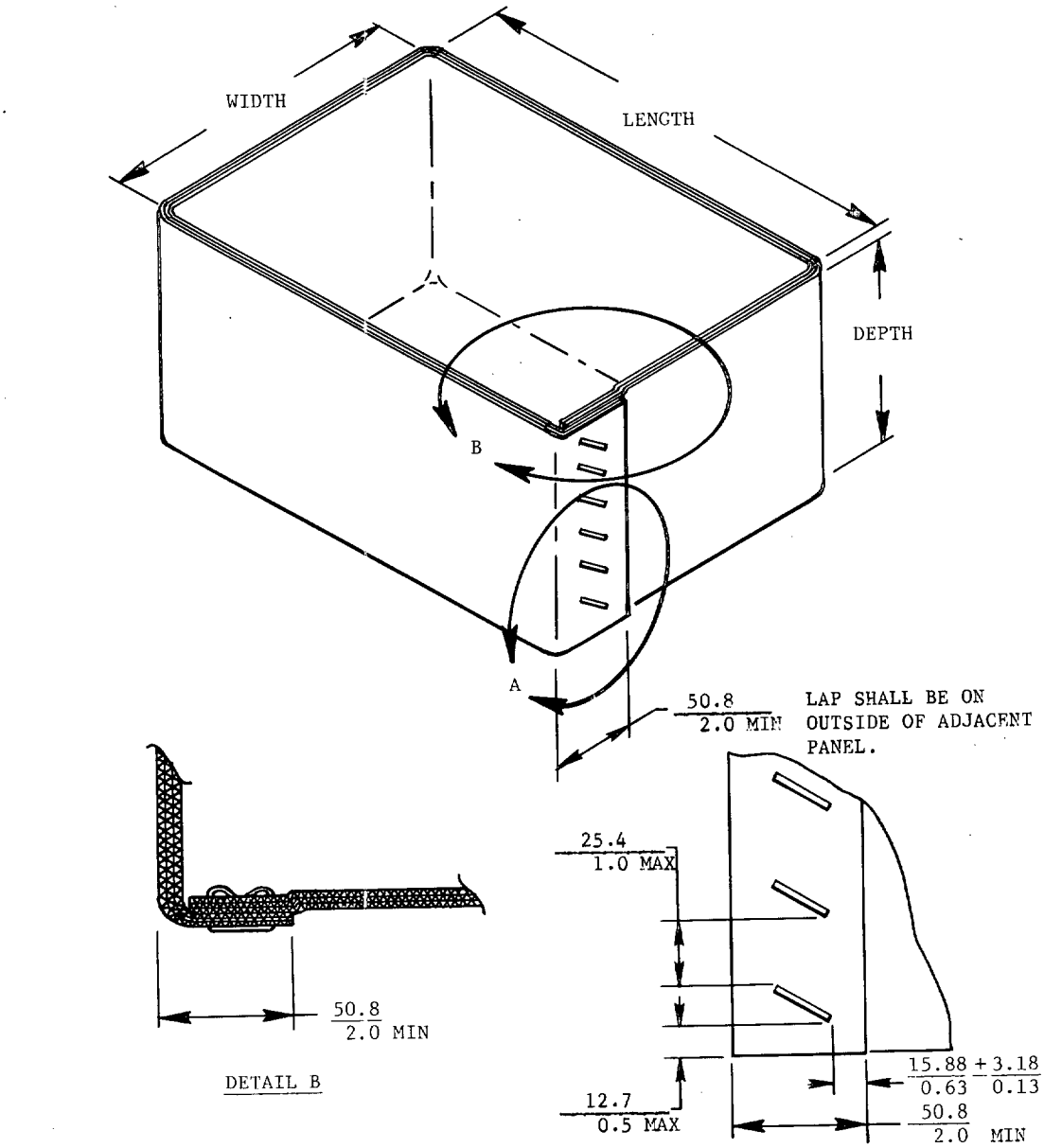
30.4 Edge protectors. Unless otherwise specified (see 6.2), edge protectors or a suitable material (see figure 8) shall be used with metallic strapping to prevent damage to the fiberboard. The protectors shall be approximately 1-3/4 inches (44.5mm) wide, and the distance between the inside edges of the protrusions shall be approximately 1 inch (25.4 mm). The indentation shall be at the center of the protector. The width of the protrusions and indentations shall be approximately 3/16 inch (9.525 mm).

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TABLE VIII. Base dimensions

SIZE OF CONTAINER	BASE SIZE O.D.		PLYWOOD THICKNESS				NUMBER BLOCKS AND LOCATION
			INCHES		MILLIMETERS		
	INCHES	MILLIMETERS	MAX	MIN	MAX	MIN	
Sixteenth and eighth	24 x 20	609.6 x 508.0	3/8	3/8	9.525	9.525	4 (1 block each corner)
Quarter	40 x 24	1016.0 x 609.6	1/2	3/8	12.700	9.525	4 (1 block each corner)
Half	48 x 40	1219.2 x 1016.0	1/2	3/8	12.700	9.525	9 (3 rows of 3 blocks equally spaced)
Full	82-1/2 x 48	2095.5 x 1219.2	1/2	3/8	12.700	9.525	15 (3 rows of 5 blocks equally spaced)

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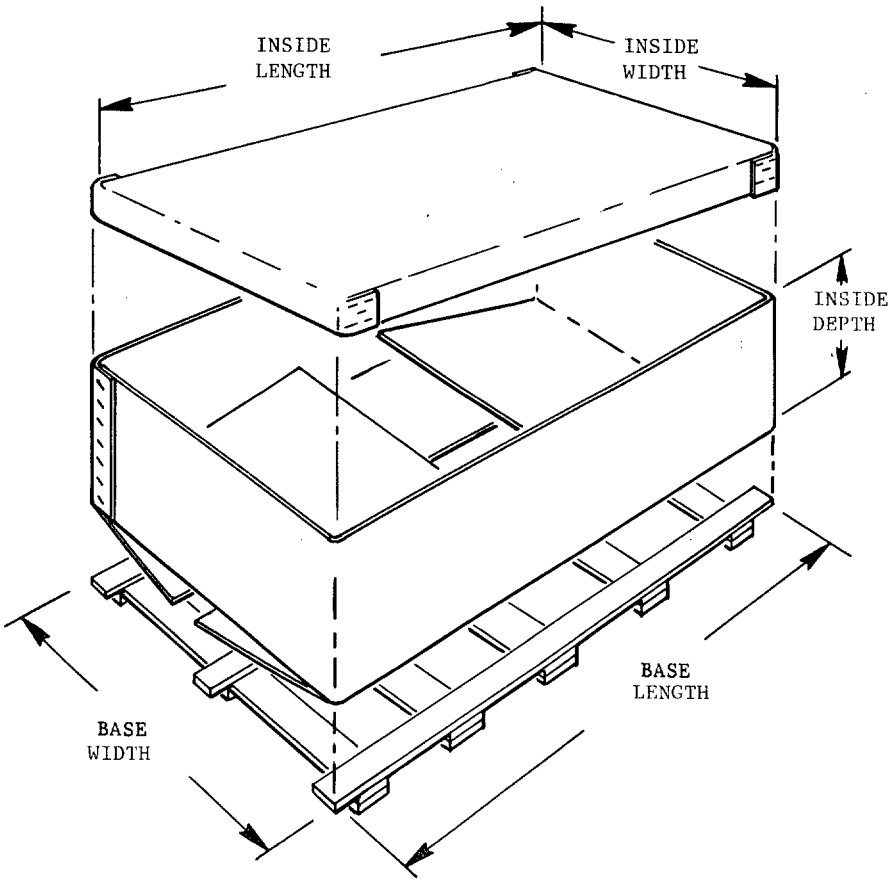


DIMENSIONS ARE IN MILLIMETERS / INCHES

SLEEVE FOR STYLE S BOX BODY

FIGURE 1

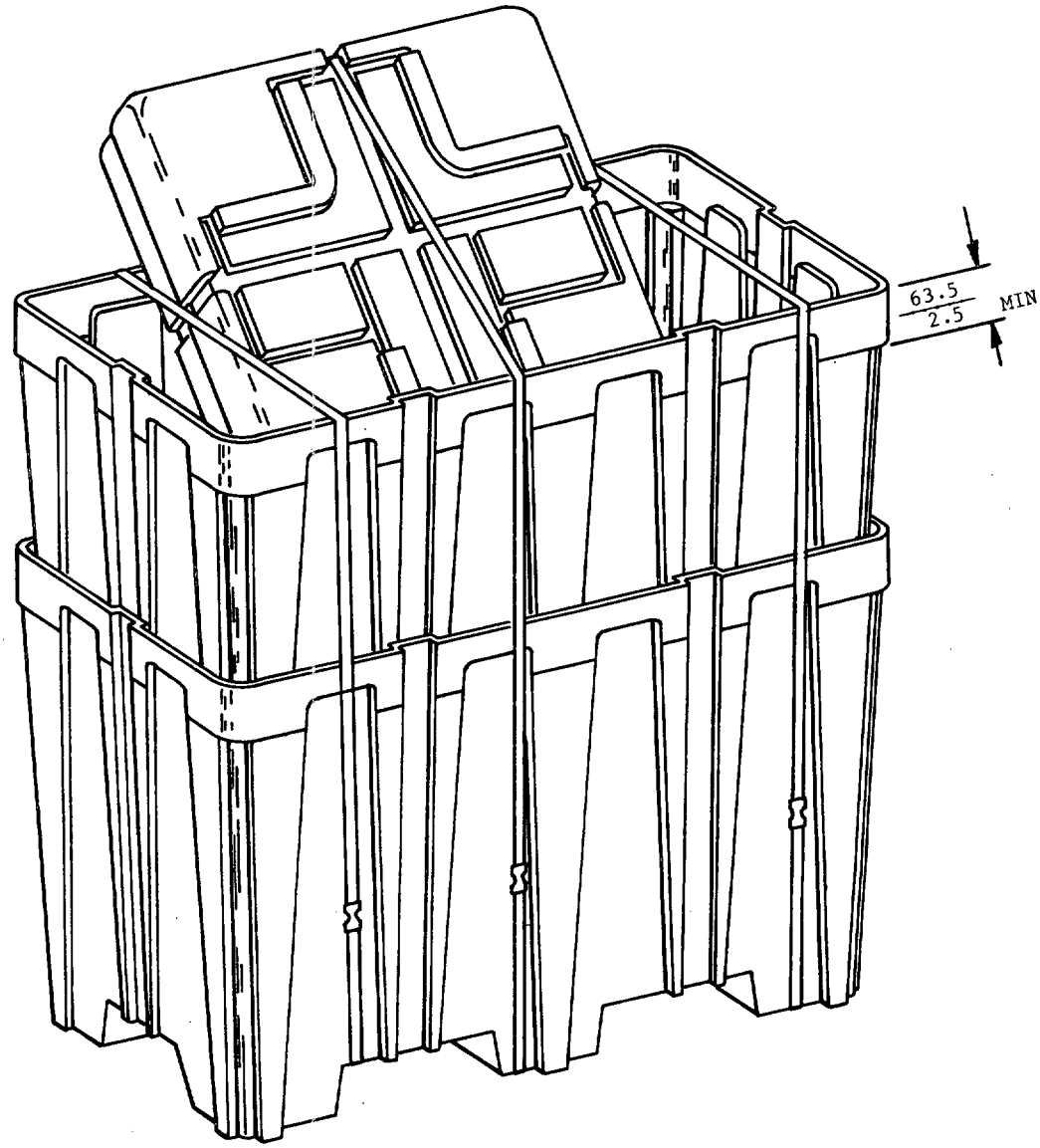
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STYLE T BOX BODY W/COVER AND BASE

FIGURE 2

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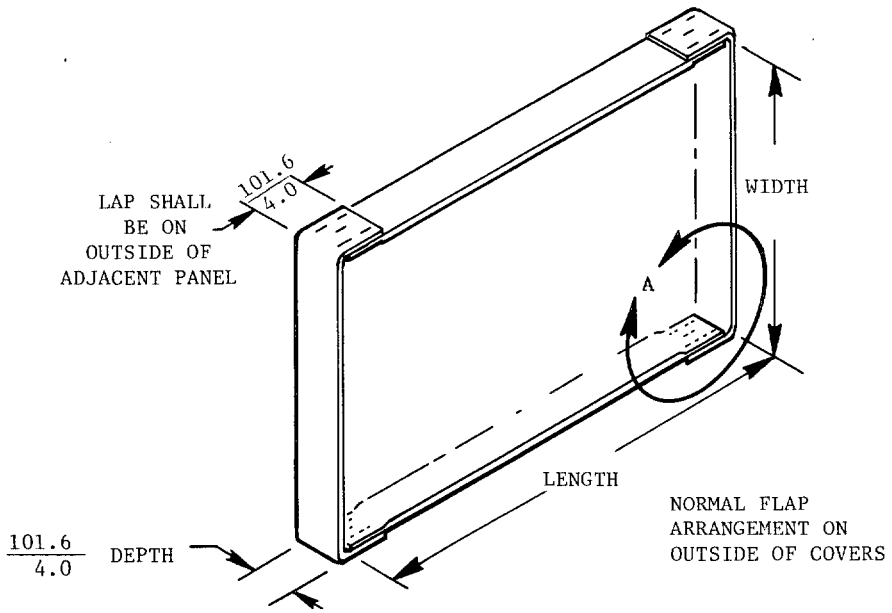


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INCHES

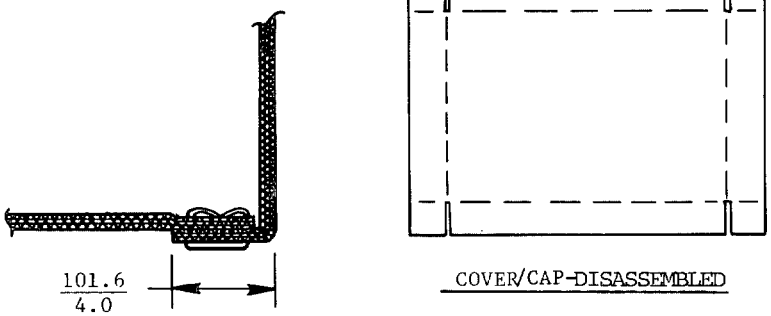
NESTED STYLE U BOX BODY AND COVER

FIGURE 3

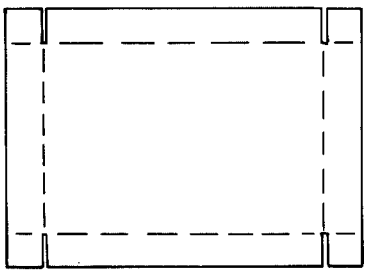
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COVER/CAP-ASSEMBLED



DETAIL A



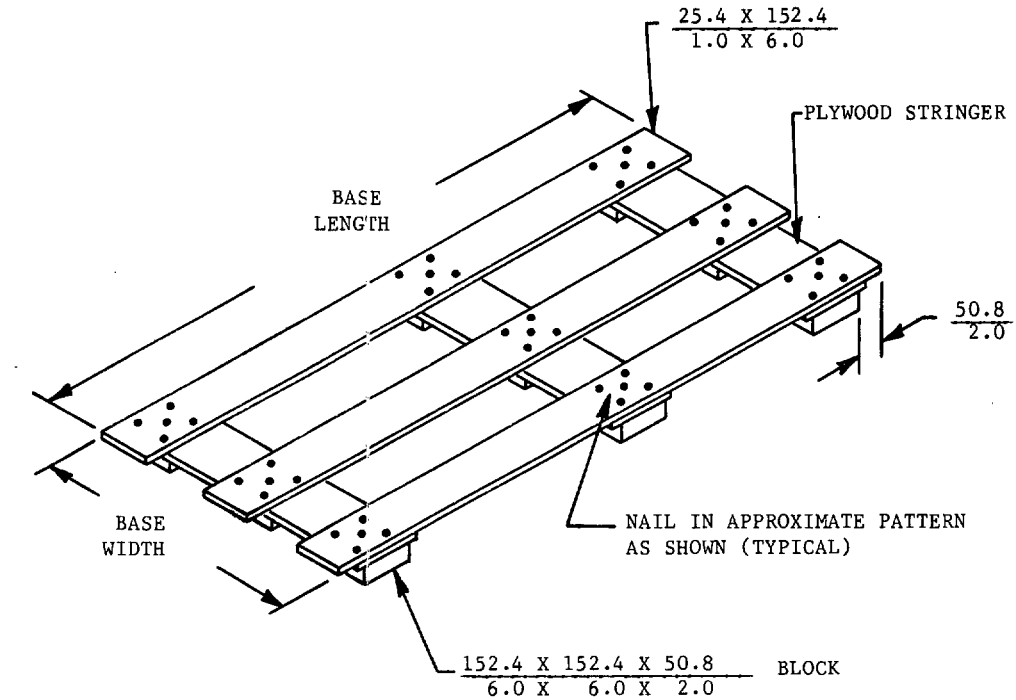
COVER/CAP-DISASSEMBLED

DIMENSIONS ARE IN MILLIMETERS
INCHES

COVERS AND CAPS FOR STYLE S AND T BOXES

FIGURE 4

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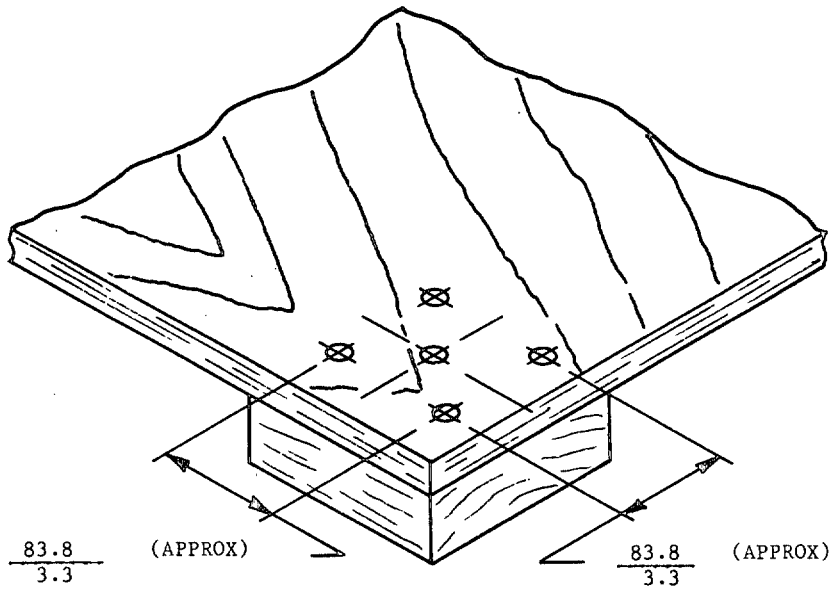


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INCHES

BASE FOR FIBERBOARD BOXES

FIGURE 5

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NAILING PATTERN TYPICAL FOR ALL
BLOCKS. 5 NAILS REQUIRED FOR
EACH BLOCK ON BASE

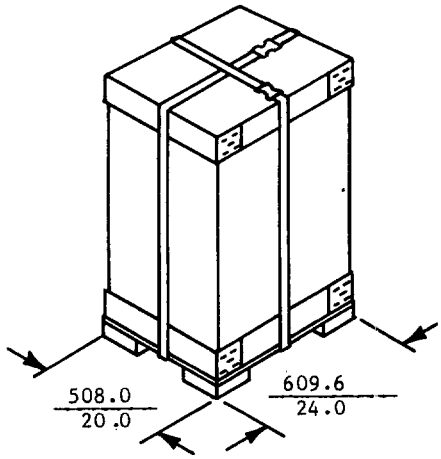
DIMENSIONS ARE IN MILLIMETERS
INCHES

BASE FOR FIBERBOARD BOXES

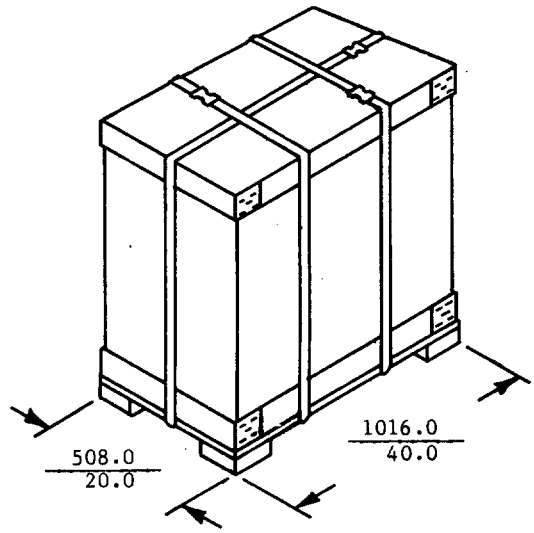
FIGURE 6

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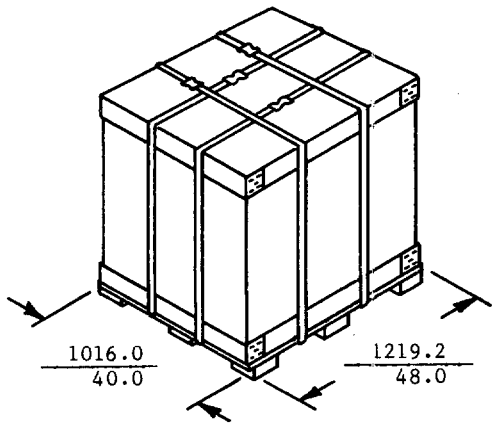
SIXTEENTH AND EIGHTH SIZE



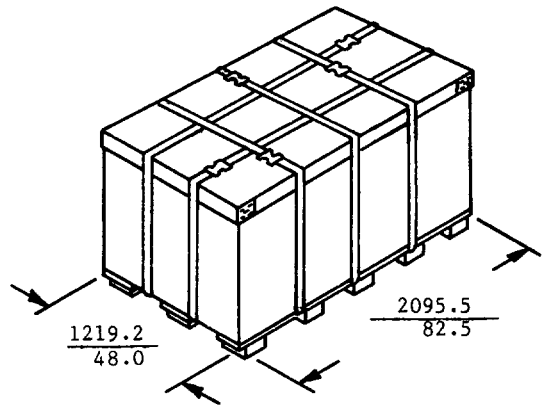
QUARTER SIZE



HALF SIZE



FULL SIZE

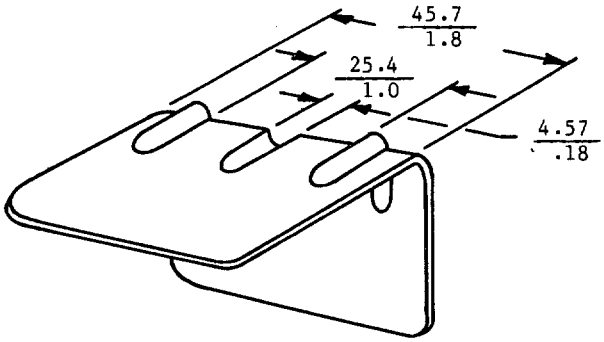


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INCHES

STRAPPING PATTERN

FIGURE 7

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EDGE PROTECTOR

FIGURE 8

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MIL-B-38721B (USAF) Boxes, Consolidation, Fiberboard and Polyolefin

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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